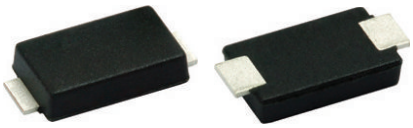
# Hyperfast Rectifier, 2 A FRED Pt®

**eSMP® Series**



Top View Bottom View

**SlimSMA HV (DO-221AC)**

Cathode ○—|◀|—○ Anode

## LINKS TO ADDITIONAL RESOURCES

3D Models

| PRIMARY CHARACTERISTICS | |
|---|---|
| $I_{F(AV)}$ | 2 A |
| $V_R$ | 1200 V |
| $V_F$ at $I_F$ | 1.6 V |
| $t_{rr}$ | 45 ns |
| $T_J$ max. | 175 °C |
| Package | SlimSMA HV (DO-221AC) |
| Circuit configuration | Single |

## FEATURES

- Minimum creepage distance 3.2 mm guaranteed by design
- Comparative Tracking Index: CTI ≥ 600
- Hyperfast recovery time, reduced $Q_{rr}$, and soft recovery
- 175 °C maximum operating junction temperature
- Specified for output and snubber operation
- Low forward voltage drop
- Low leakage current
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE

## DESCRIPTION / APPLICATIONS

State of the art hyperfast recovery rectifiers specifically designed with optimized performance of forward voltage drop and hyperfast recovery time.

The planar structure and the platinum doped life time control guarantee the best overall performance, ruggedness, and reliability characteristics.

These devices are intended for use as clamp, snubber and freewheeling diode in a flyback aux power supplies, bootstrap and desaturate for HV MOSFET and IGBT driver, high frequency rectifiers in a cuk and sepic circuit for LED lighting.

Their extremely optimized stored charge and low recovery current minimize the switching losses and reduce power dissipation in the switching element.

## MECHANICAL DATA

**Case:** SlimSMA HV (DO-221AC)

Molding compound meets UL 94 V-0 flammability rating

**Terminals:** matte tin plated leads, solderable per J-STD-002

**Polarity:** color band denotes cathode end

| ABSOLUTE MAXIMUM RATINGS | | | | |
|---|---|---|---|---|
| PARAMETER | SYMBOL | TEST CONDITIONS | VALUES | UNITS |
| Peak repetitive reverse voltage | $V_{RRM}$ | | 1200 | V |
| Average rectified forward current | $I_{F(AV)}$ | $T_{Sp}$ = 95 °C, DC conduction | 2 | A |
| Non-repetitive peak surge current | $I_{FSM}$ | $T_J$ = 25 °C, 8.3 ms sine pulse | 21 | |
| Operating junction and storage temperatures | $T_J$, $T_{Stg}$ | | -55 to +175 | °C |



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| ELECTRICAL SPECIFICATIONS ($T_J$ = 25 °C unless otherwise specified) | | | | | | |
|---|---|---|---|---|---|---|
| PARAMETER | SYMBOL | TEST CONDITIONS | MIN. | TYP. | MAX. | UNITS |
| Breakdown voltage, blocking voltage | $V_{BR}$, $V_R$ | $I_R$ = 100 µA | 1200 | - | - | V |
| Forward voltage drop | $V_F$ | $I_F$ = 2 A | - | 2.0 | 2.50 | V |
| | | $I_F$ = 2 A, $T_J$ = 125 °C | - | 1.7 | 1.95 | |
| | | $I_F$ = 2 A, $T_J$ = 150 °C | - | 1.6 | 1.85 | |
| Reverse leakage current | $I_R$ | $V_R$ = $V_R$ rated | - | - | 3 | µA |
| | | $T_J$ = 125 °C, $V_R$ = $V_R$ rated | - | - | 30 | |
| Junction capacitance | $C_T$ | $V_R$ = 1200 V, 1 MHz | - | 3.0 | - | pF |
| | | $V_R$ = 4 V, 1 MHz | - | 7.0 | - | |

| DYNAMIC RECOVERY CHARACTERISTICS ($T_J$ = 25 °C unless otherwise specified) | | | | | | | |
|---|---|---|---|---|---|---|---|
| PARAMETER | SYMBOL | TEST CONDITIONS | | MIN. | TYP. | MAX. | UNITS |
| Reverse recovery time | $t_{rr}$ | $I_F$ = 0.5 A, $I_R$ = 1 A, $I_{rr}$ = 0.25 A | | - | 35 | 45 | ns |
| | | $T_J$ = 25 °C | $I_F$ = 2 A, $dI_F/dt$ = 200 A/µs, $V_R$ = 800 V | - | 105 | - | |
| | | $T_J$ = 125 °C | | - | 145 | - | |
| Peak recovery current | $I_{RRM}$ | $T_J$ = 25 °C | | - | 4.0 | - | A |
| | | $T_J$ = 125 °C | | - | 5.0 | - | |
| Reverse recovery charge | $Q_{rr}$ | $T_J$ = 25 °C | | - | 165 | - | nC |
| | | $T_J$ = 125 °C | | - | 315 | - | |

| THERMAL AND MECHANICAL SPECIFICATIONS | | | | | | |
|---|---|---|---|---|---|---|
| PARAMETER | SYMBOL | TEST CONDITIONS | MIN. | TYP. | MAX. | UNITS |
| Maximum junction and storage temperature range | $T_J$, $T_{Stg}$ | | -55 | - | 175 | °C |
| Thermal resistance, junction to mount | $R_{thJM}$ (1) | Device mounted on PCB with 2 x 3.5 mm soldering lands | - | 18 | 22 | °C/W |
| Thermal resistance, junction to ambient | $R_{thJA}$ | Device mounted on PCB with recommended pad size | - | 120 | - | °C/W |
| Approximate weight | | | 0.032 | | | g |
| Marking device | | Case style SlimSMA HV (DO-221AC) | 2X12 | | | |

**Note**

(1) Thermal resistance junction to mount follows JEDEC® 51-14 transient dual interface test method (TDIM)

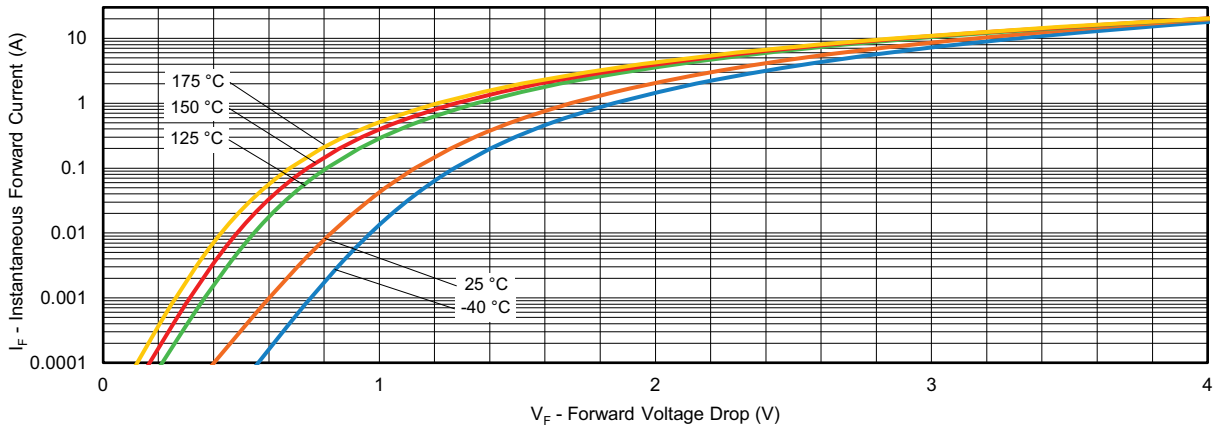



Fig. 1 - Typical Forward Voltage Drop Characteristics


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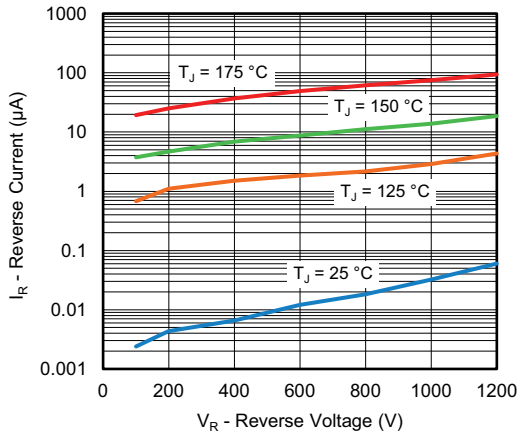



Fig. 2 - Typical Values of Reverse Current vs. Reverse Voltage

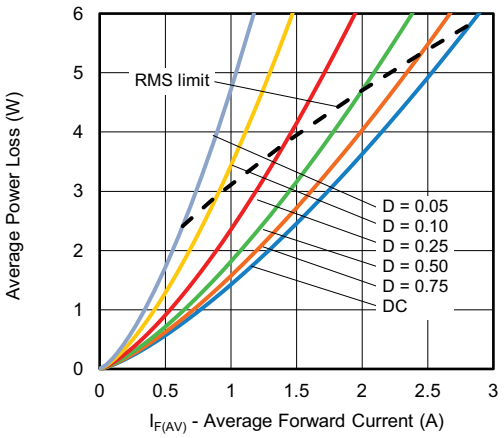



Fig. 5 - Forward Power Loss Characteristics

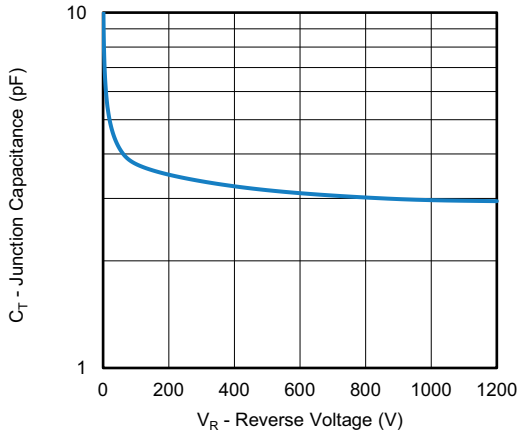



Fig. 3 - Typical Junction Capacitance vs. Reverse Voltage

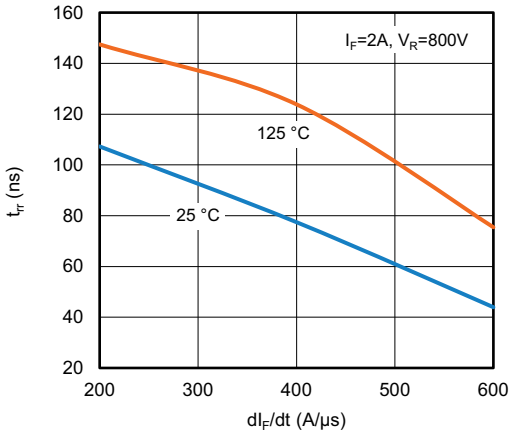



Fig. 6 - Typical Reverse Recovery Time vs. $dI_F/dt$

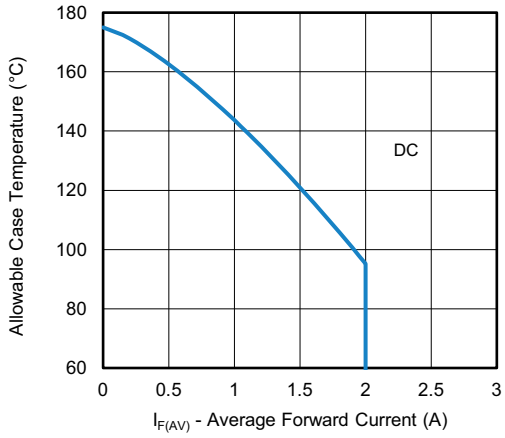



Fig. 4 - Maximum Allowable Case Temperature vs. Average Forward Current

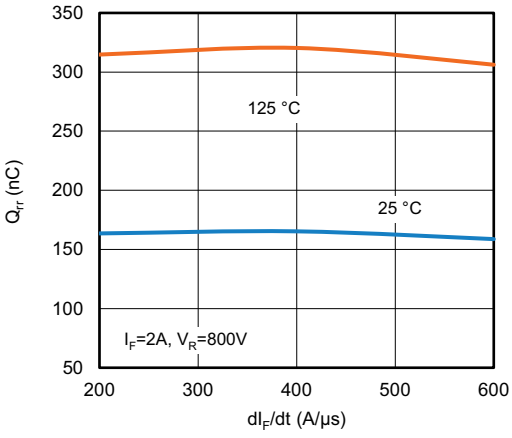



Fig. 7 - Typical Stored Charge vs. $dI_F/dt$


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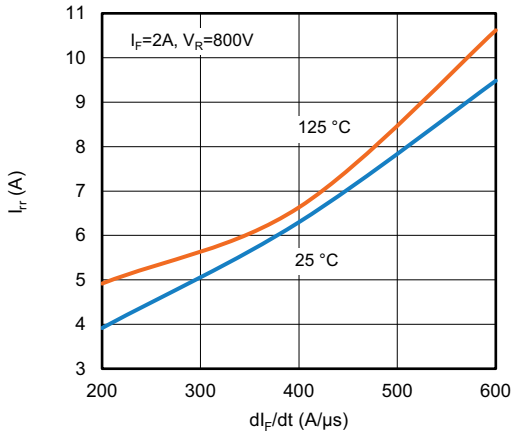



Fig. 8 - $I_{rr}$ (A) vs. $dI_F/dt$

**Note**

(1) Formula used: $T_C = T_J - (Pd + Pd_{REV}) \times R_{thJC}$;
Pd = forward power loss = $I_{F(AV)} \times V_{FM}$ at ($I_{F(AV)}$/D) (see fig. 5);
$Pd_{REV}$ = inverse power loss = $V_{R1} \times I_R$ (1 - D); $I_R$ at $V_{R1}$ = rated $V_R$

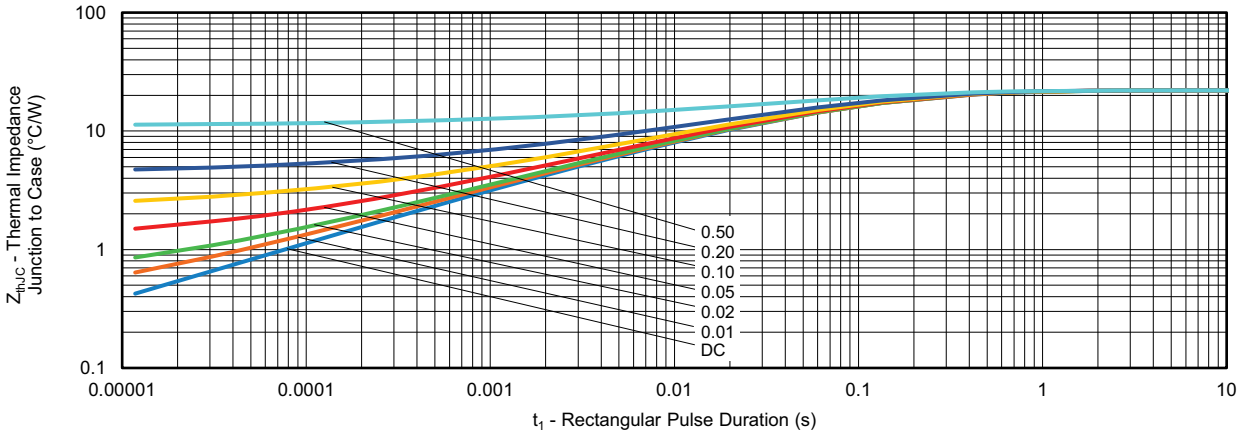
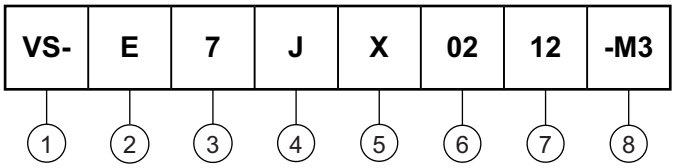



Fig. 9 - Transient Thermal Impedance, Junction to Case


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## ORDERING INFORMATION TABLE

Device code




1 - Vishay Semiconductors product

2 - Circuit configuration:
E = single diode

3 - 7 = FRED generation 7

4 - J = SlimSMA package

5 - Process type,
X = hyperfast recovery

6 - Current rating (02 = 2 A)

7 - Voltage code (12 = 1200 V)

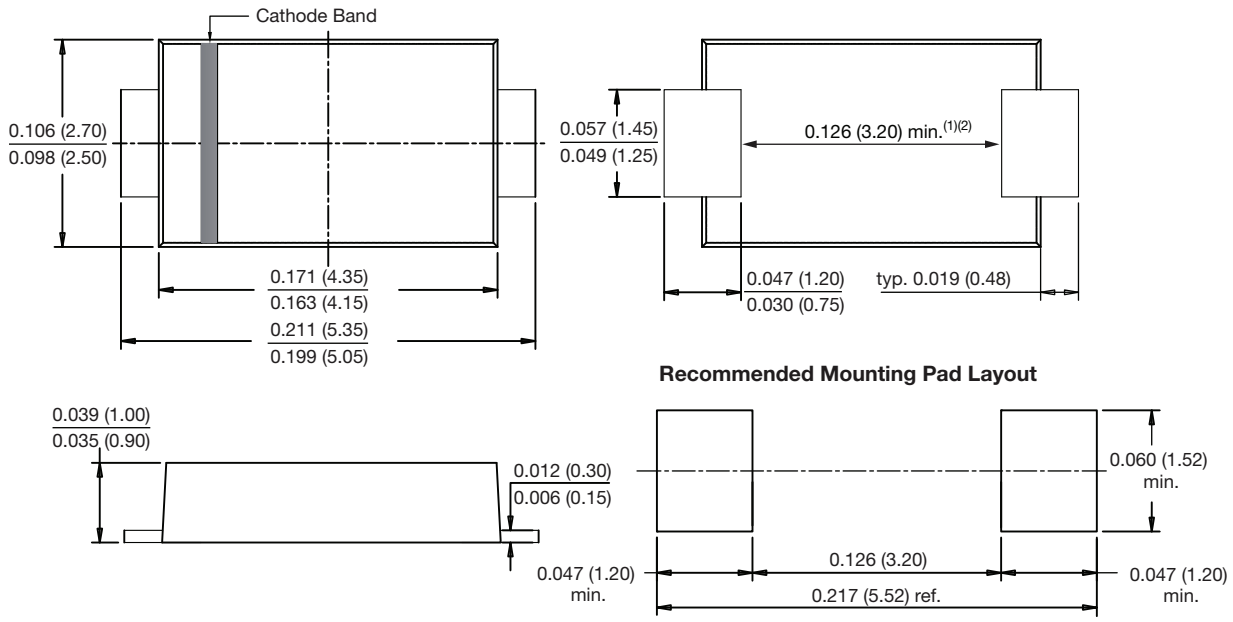
8 - M3 = halogen-free, RoHS-compliant, and terminations lead (Pb)-free

| **ORDERING INFORMATION** (Example) | | | |
|---|---|---|---|
| **PREFERRED P/N** | **QUANTITY PER REEL** | **BASE QUANTITY** | **PACKAGING DESCRIPTION** |
| VS-E7JX0212-M3/H | 3500 | 3500 | 7"diameter plastic tape and reel |
| VS-E7JX0212-M3/I | 14 000 | 14 000 | 13"diameter plastic tape and reel |

| **LINKS TO RELATED DOCUMENTS** | |
|---|---|
| Dimensions | www.vishay.com/doc?97278 |
| Part marking information | www.vishay.com/doc?95562 |
| Packaging information | www.vishay.com/doc?88869 |
| SPICE model | www.vishay.com/doc?97549 |


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**DIMENSIONS** in inches (millimeters)




**Notes**

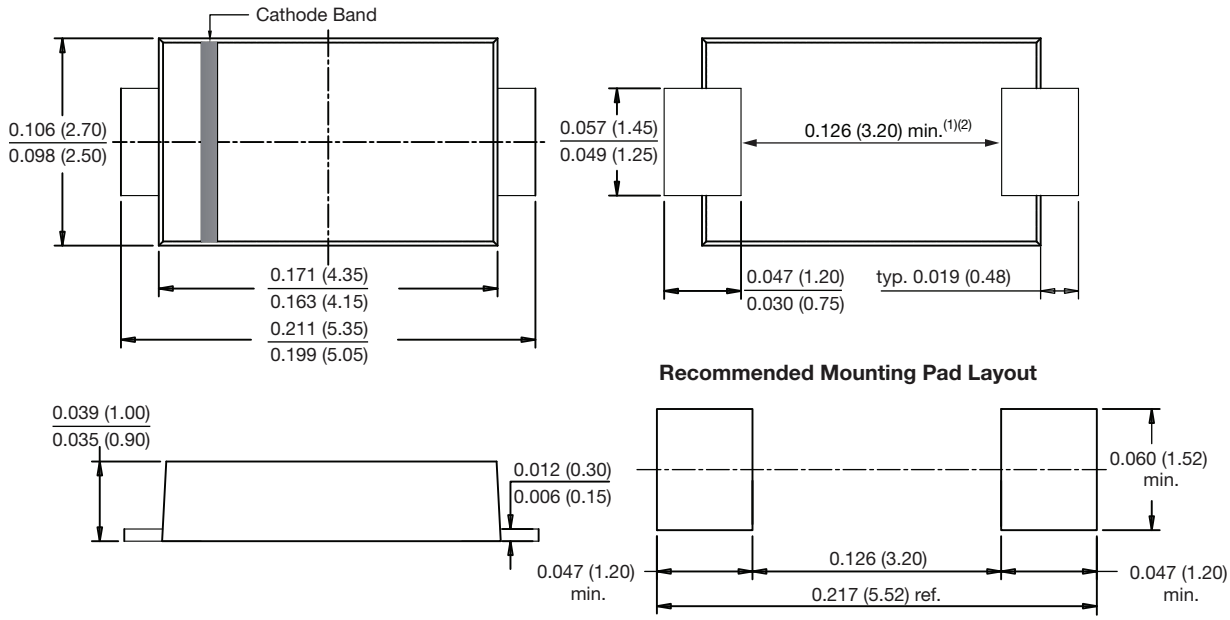
(1) Minimum creepage distance is defined and guaranteed by design

(2) For high voltage applications, end users should consider the relevant guidelines and normative on creepage and clearance distances between device terminals and PCB pads.


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# SlimSMA HV (DO-221AC)

**DIMENSIONS** in inches (millimeters)




**Notes**

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